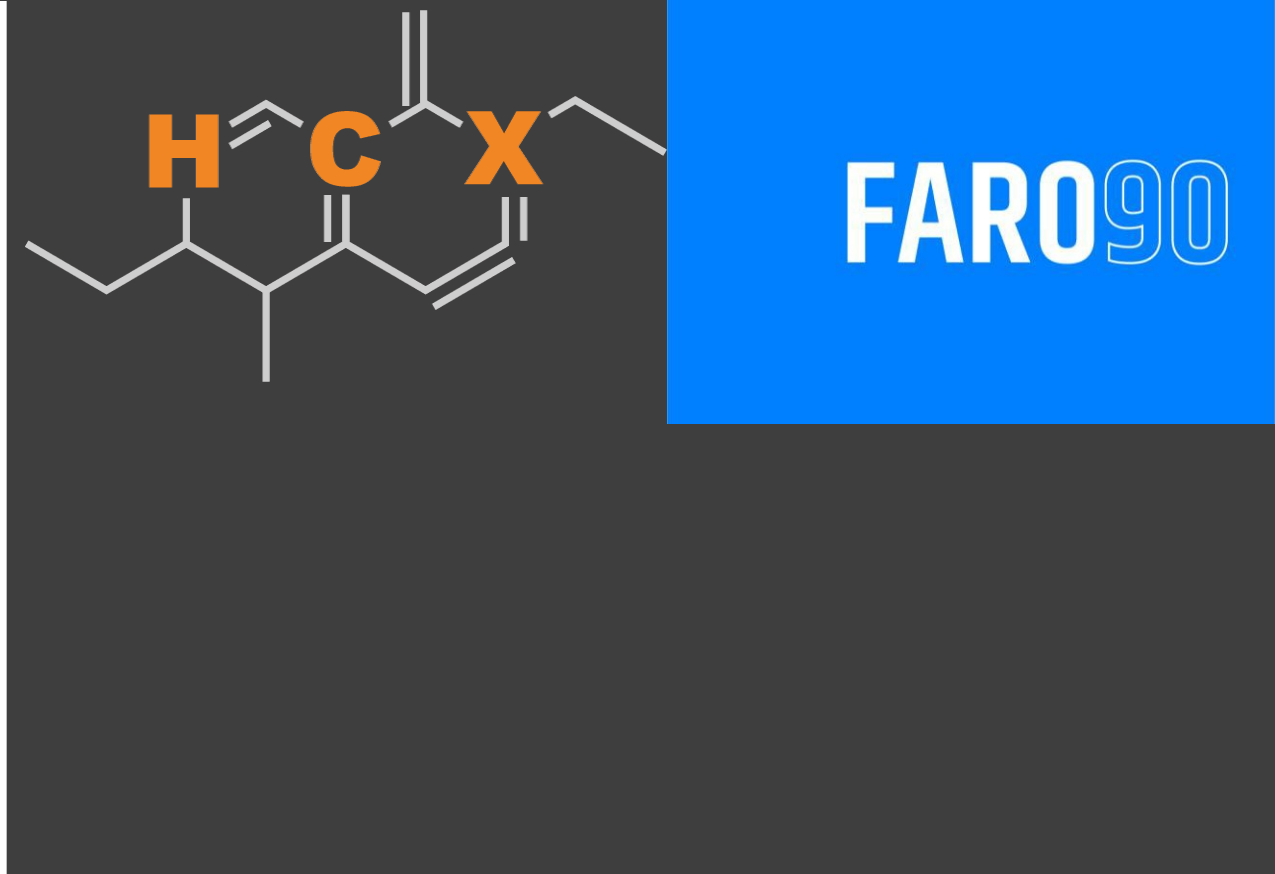




Ethanol Blending in Gasoline - Slovakia

September, 2025

Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Slovakia



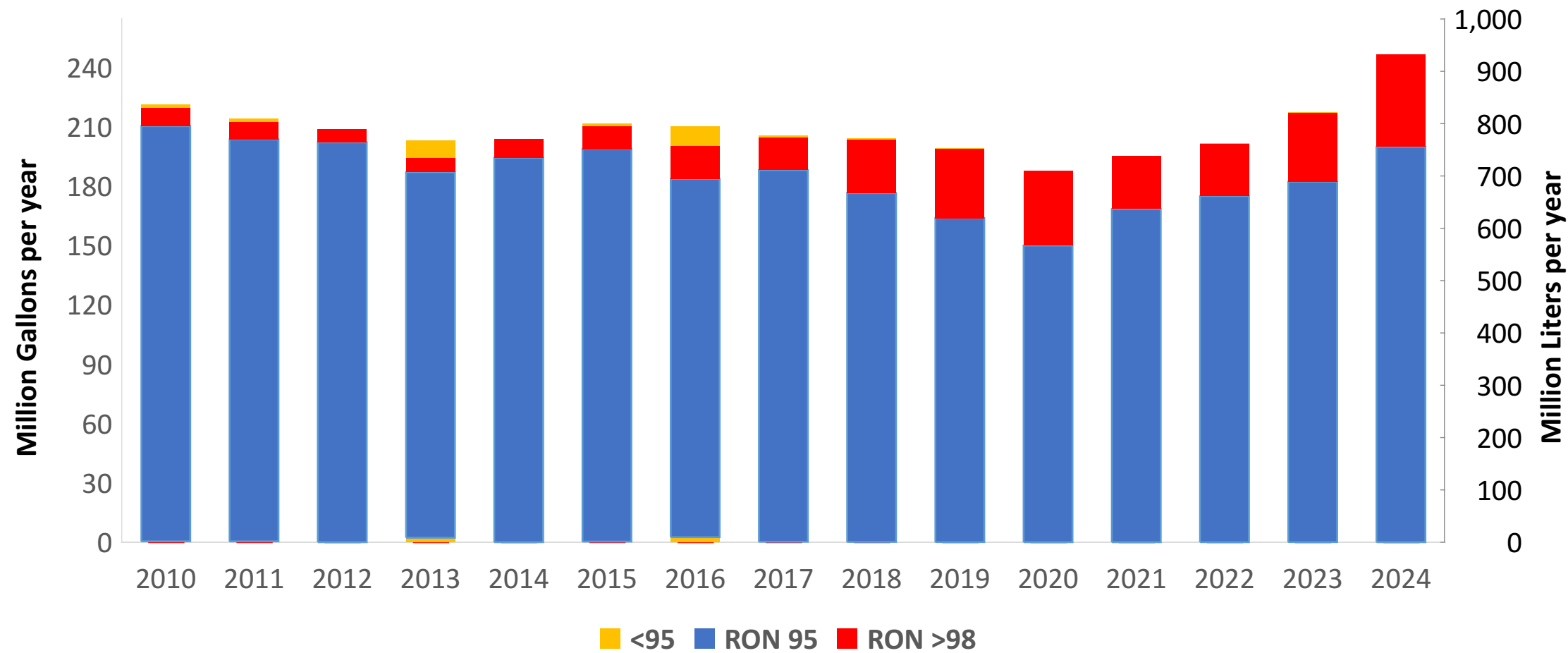
In 2024, gasoline consumption in Slovakia was 931 million liters (246 million gallons) and growth rate was 4.4%. Gasoline production and net imports were 931 million liters (and 246 million gallons) correspondingly. Slovakia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.6 RON.

In 2024, ethanol consumption in Slovakia was 80 million liters (21 million gallons) representing 8.6% of gasoline demand. Ethanol production and Net Imports were 80 and million liters (21 and million gallons) correspondingly. Ethanol source is Cereals 100%.

Source: Eurostat, European Commission

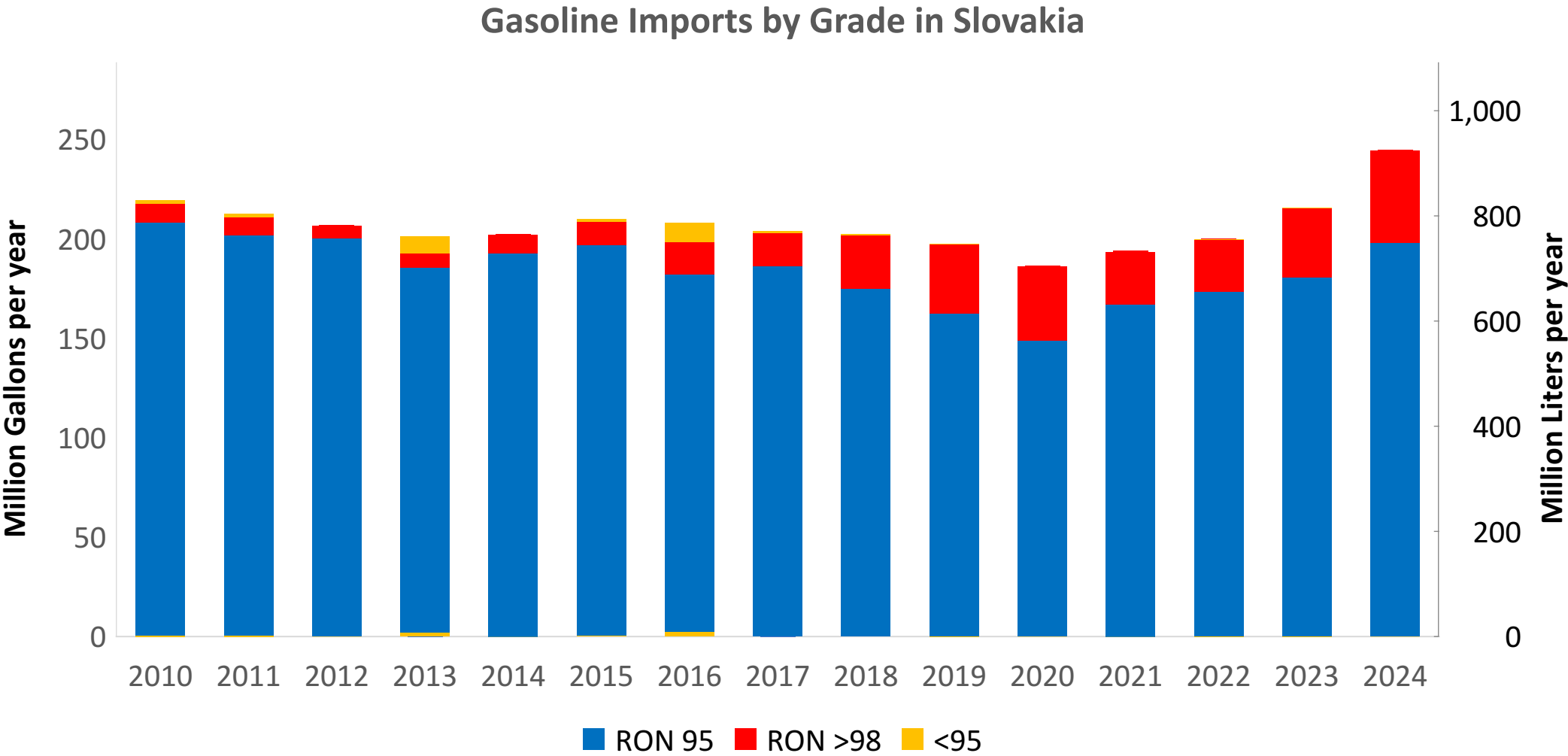
Gasoline Demand in Slovakia

Gasoline Demand by Grade in Slovakia



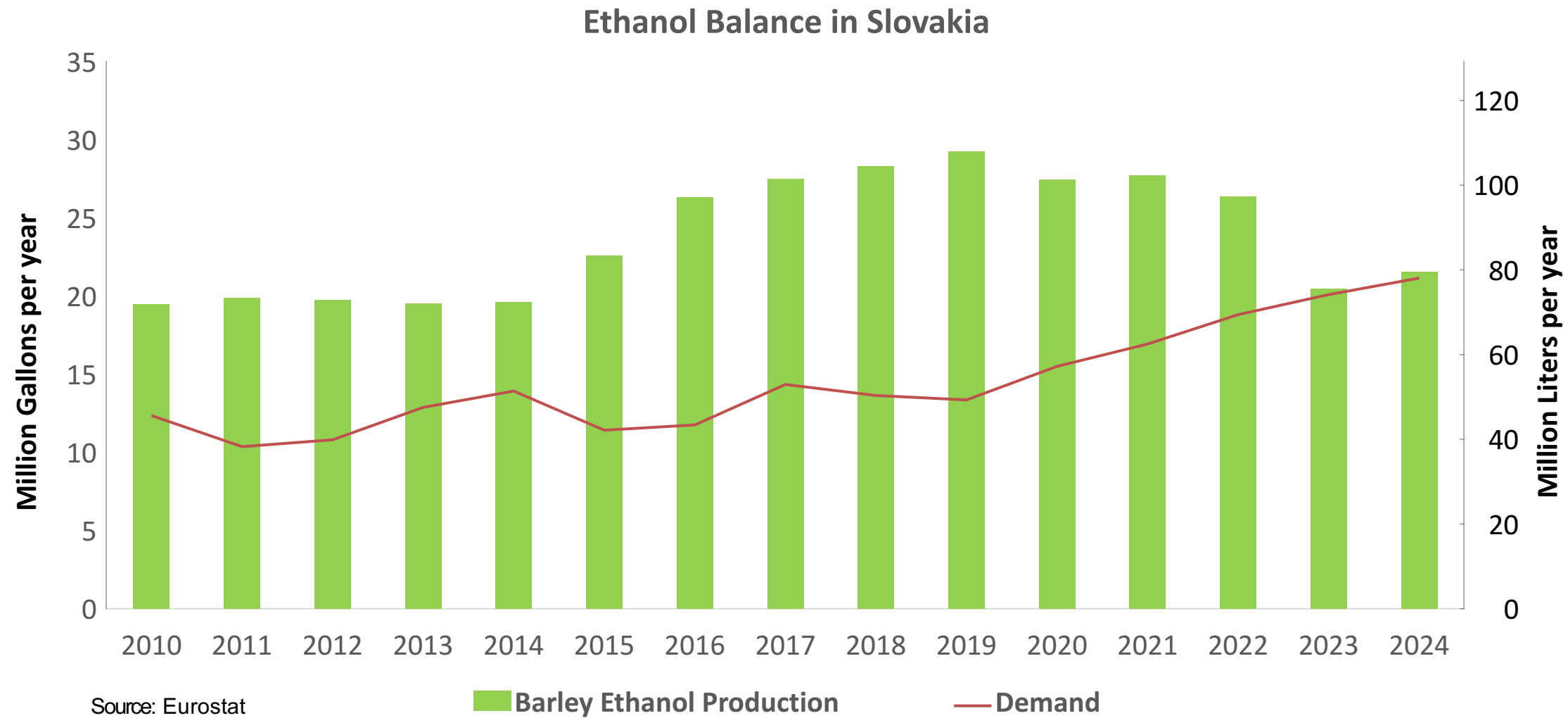
Source: Eurostat

Gasoline Imports in Slovakia



Source: Eurostat

Ethanol Balance in Slovakia



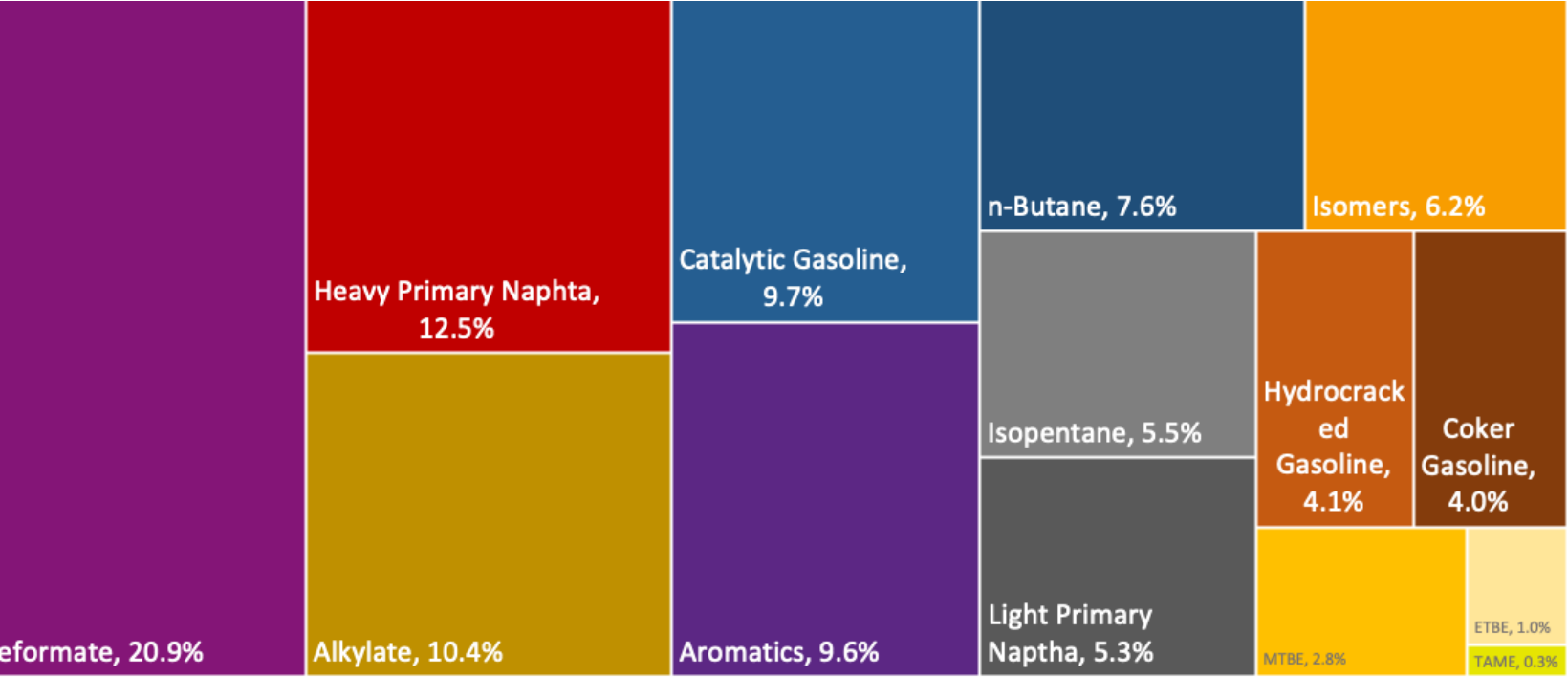
Gasoline Quality in Slovakia

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	9.2							
Advanced Biofuels (%cal)	1.1							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

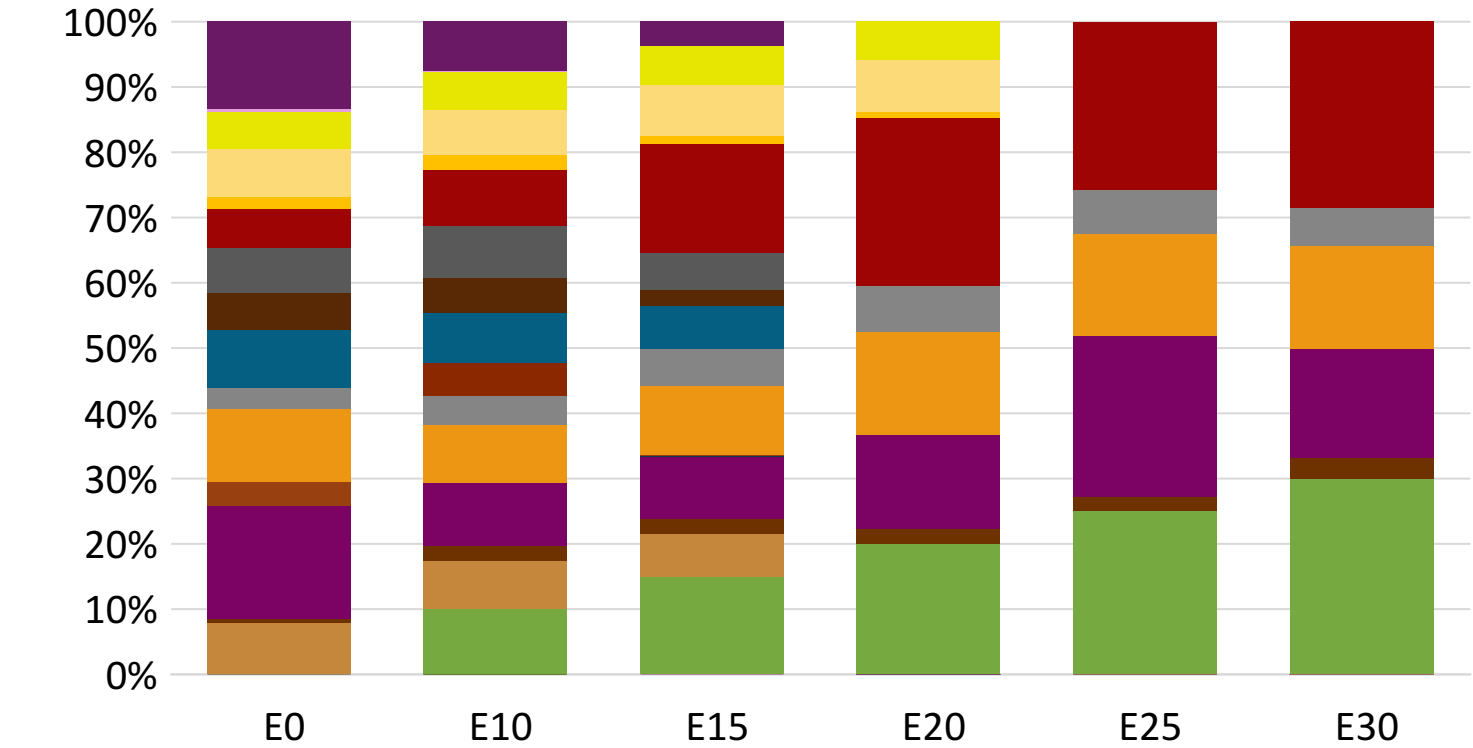
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Slovakia Available Blending Components



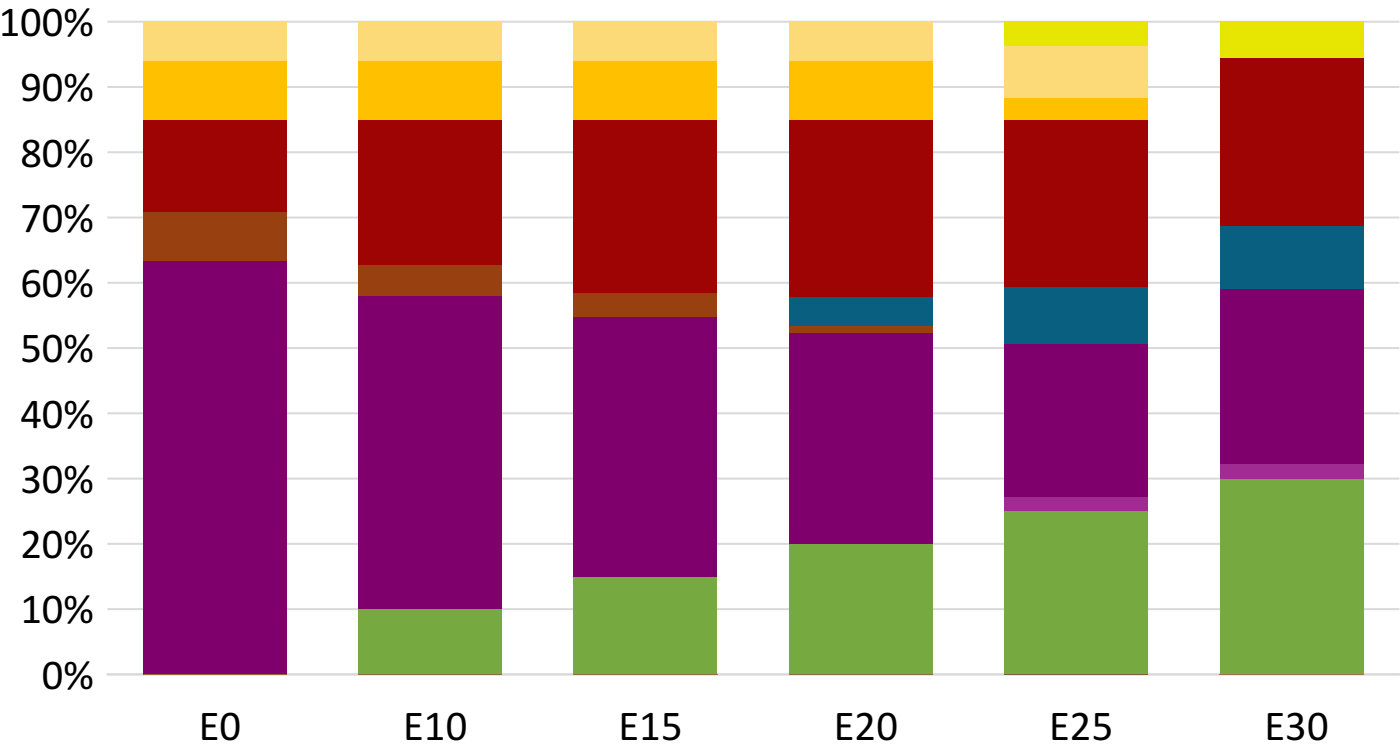
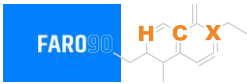
Slovakia – Gasoline 95 – Constant Octane



Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.304	\$ 2.166	\$ 2.066	\$ 1.951	\$ 1.905	\$ 1.852

Slovakia - Gasoline 98 - Constant Octane

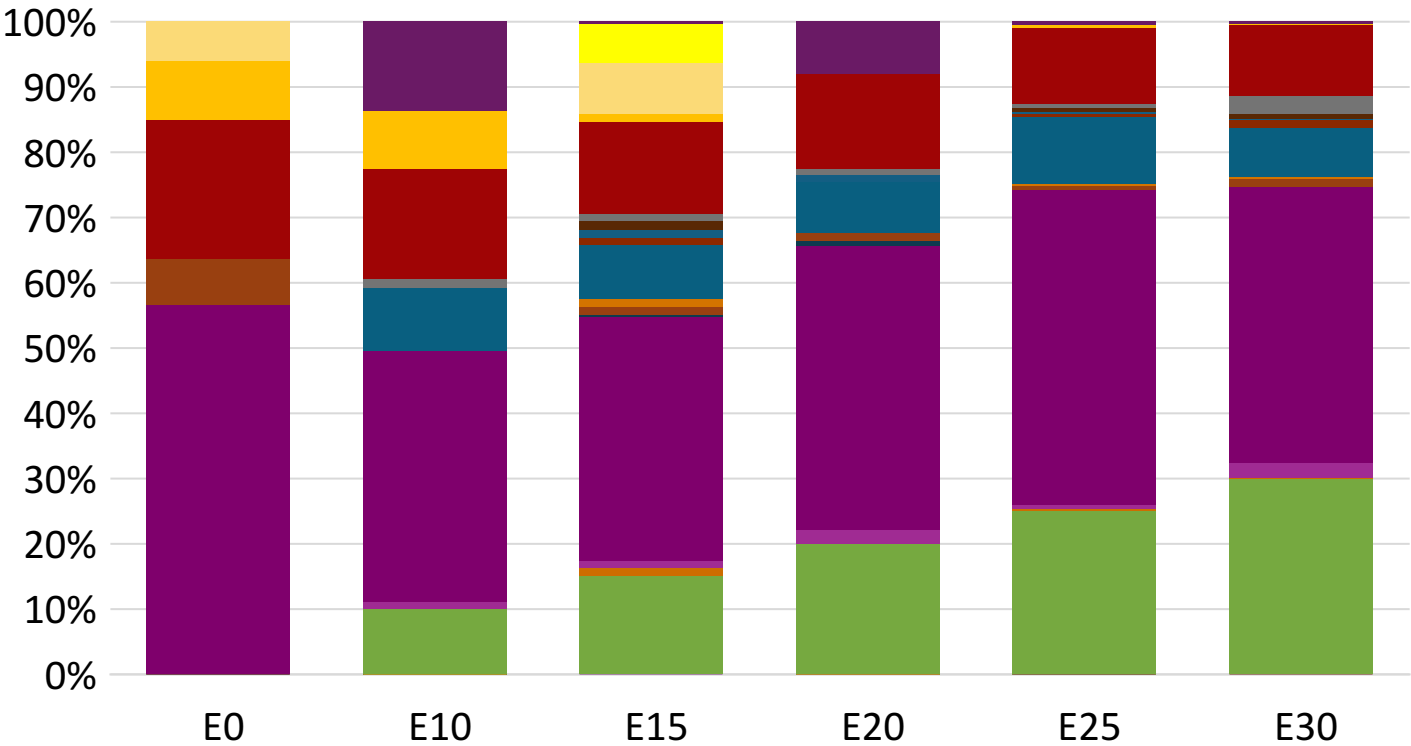


Average CIF 2024 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.324	\$ 2.207	\$ 2.138	\$ 2.066	\$ 2.001	\$ 1.951

Source: Faro90

Slovakia – Gasoline 95 – Increased Octane

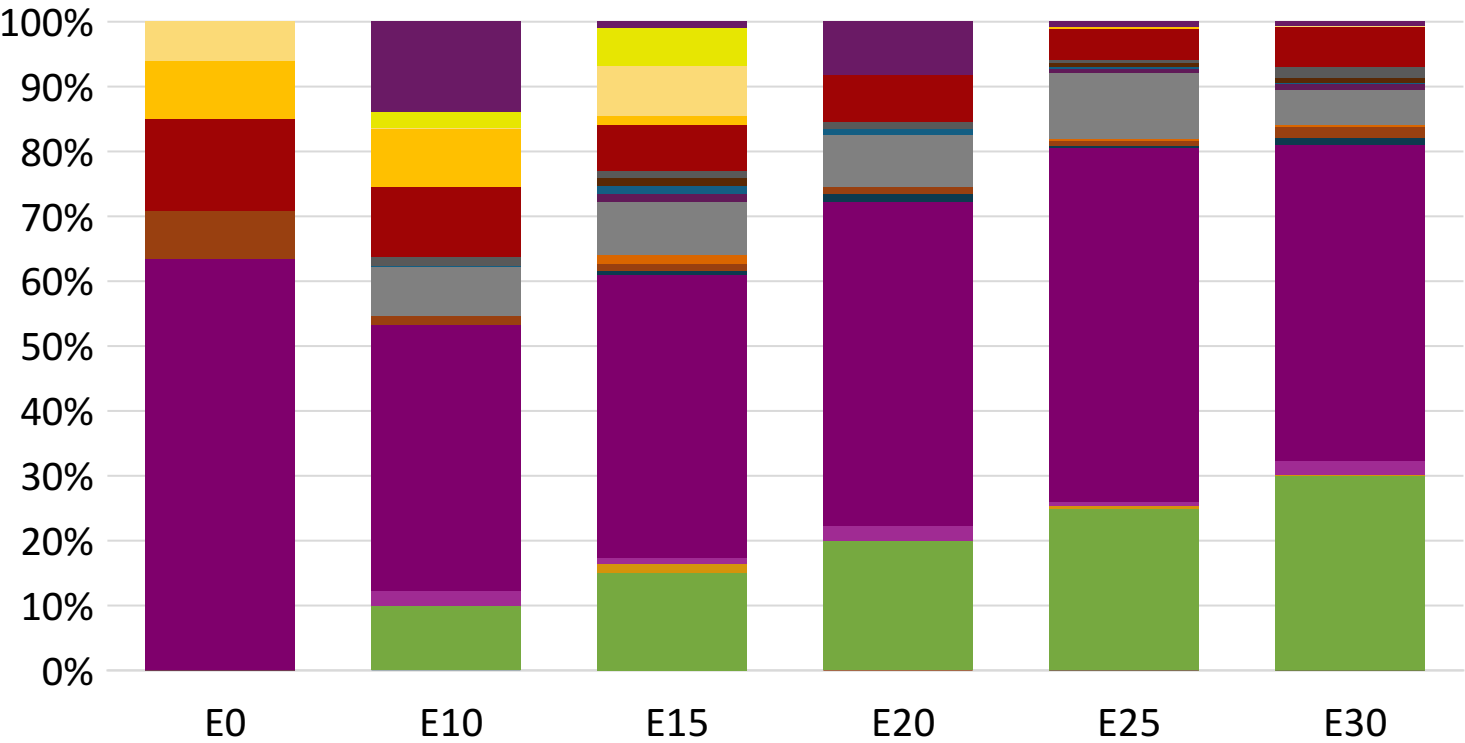


Average CIF 2024 Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 2.192	\$ 2.192	\$ 2.192	\$ 2.192	\$ 2.192	\$ 2.192

Source: Faro90

Slovakia – Gasoline 98 – Increased Octane



Average CIF 2024 Prices

Octane (RON)		98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$	2.324	\$ 2.324	\$ 2.324	\$ 2.324	\$ 2.324	\$ 2.324

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

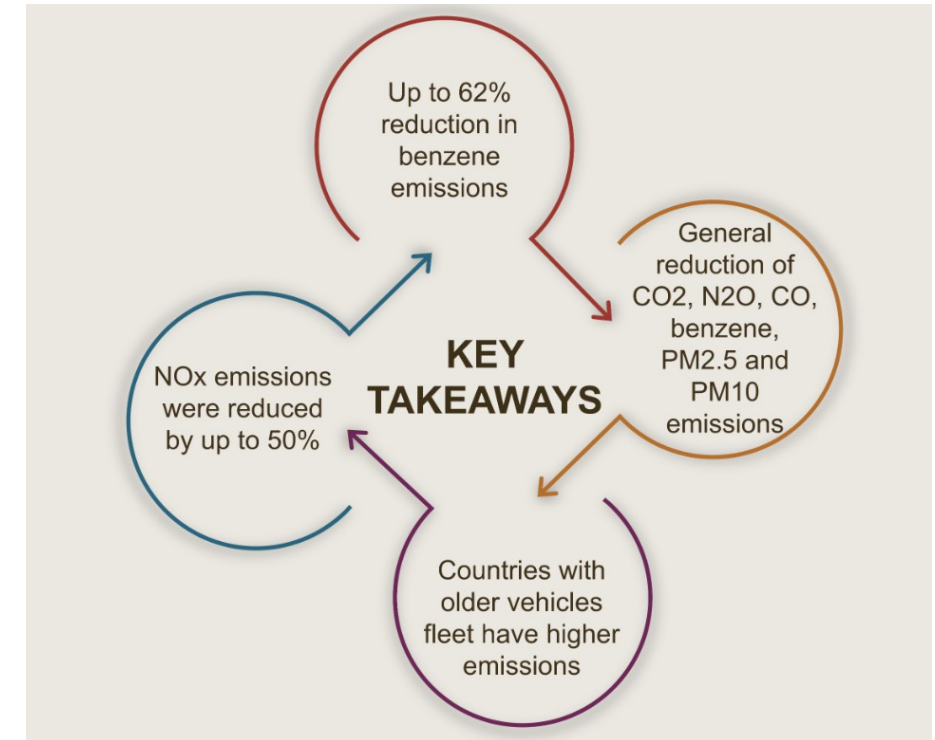
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

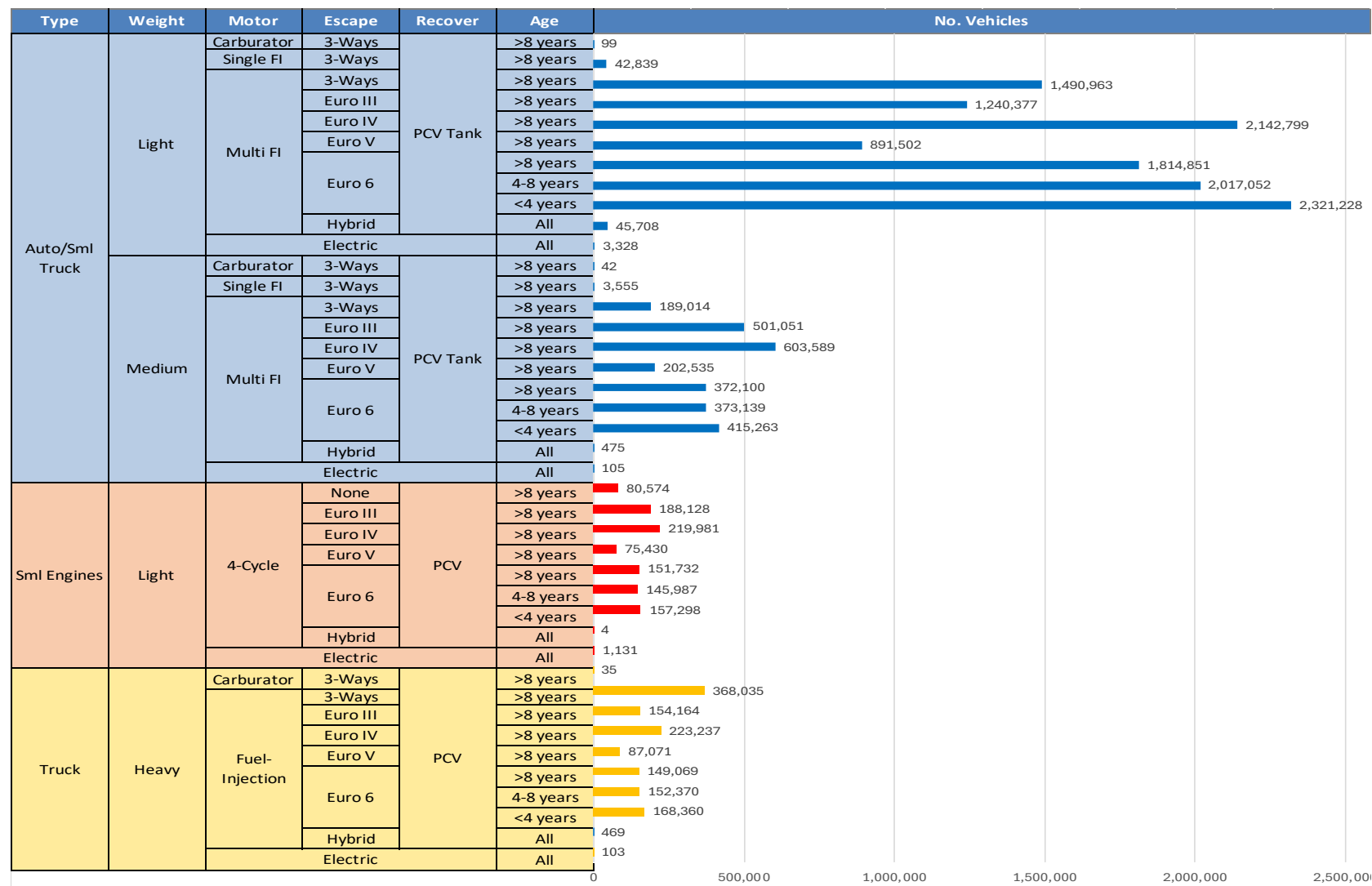
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



*Source: Bureau of transportation statistics.

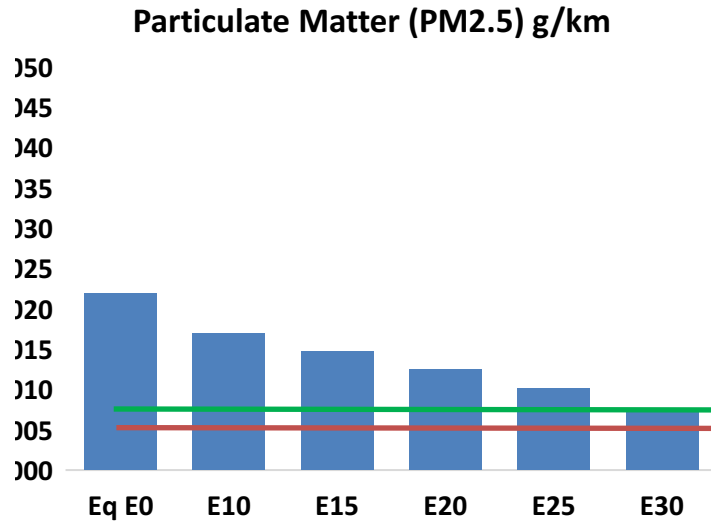
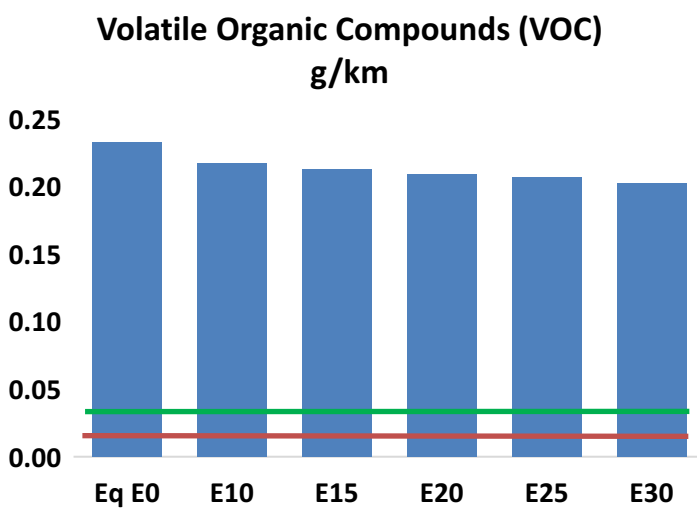
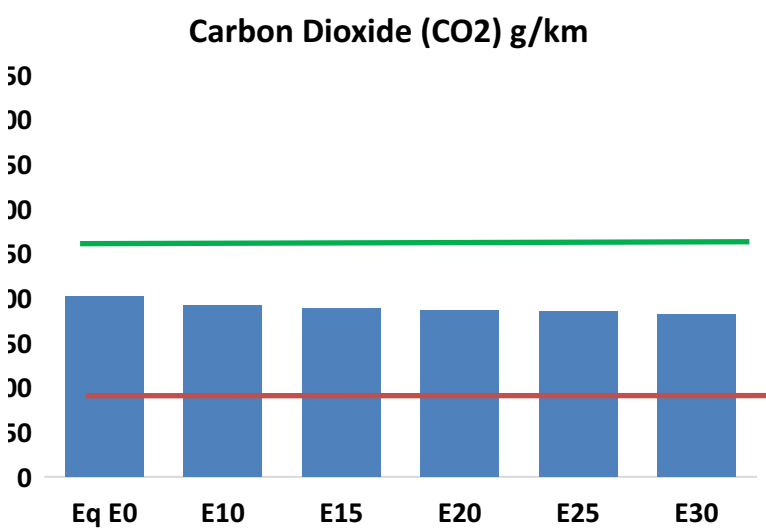
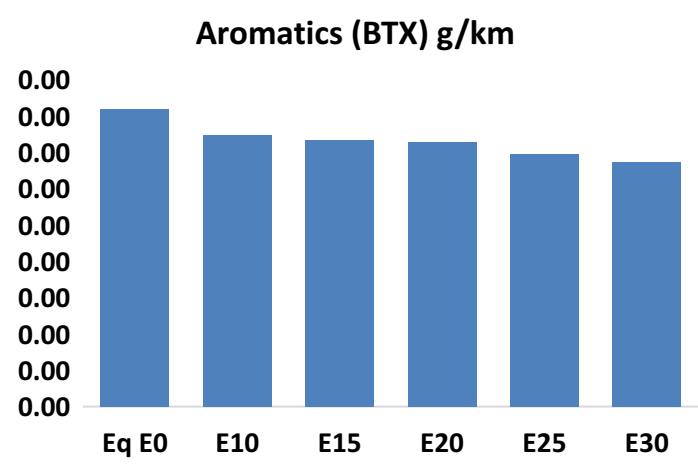
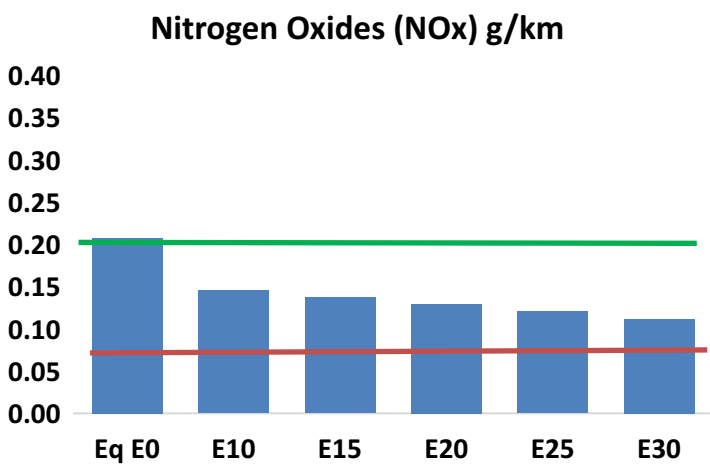
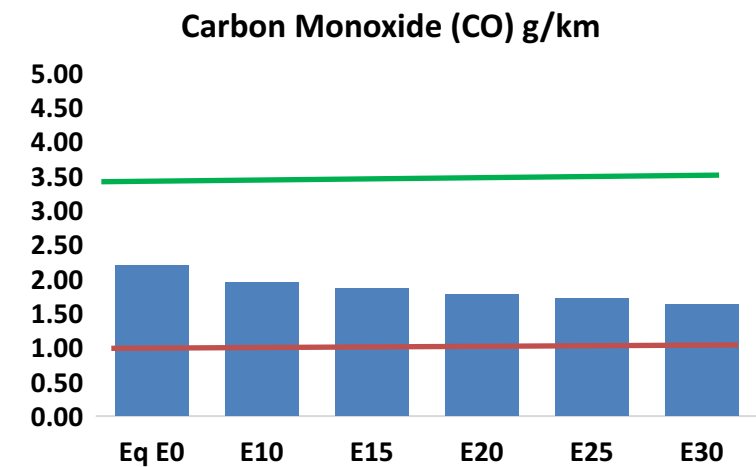
Gasoline Vehicle Fleet - Slovakia



Vehicle Fleet: 5,792,410
Gasoline Vehicle Fleet: 1,843,374

Source: Eurostat, ACEA

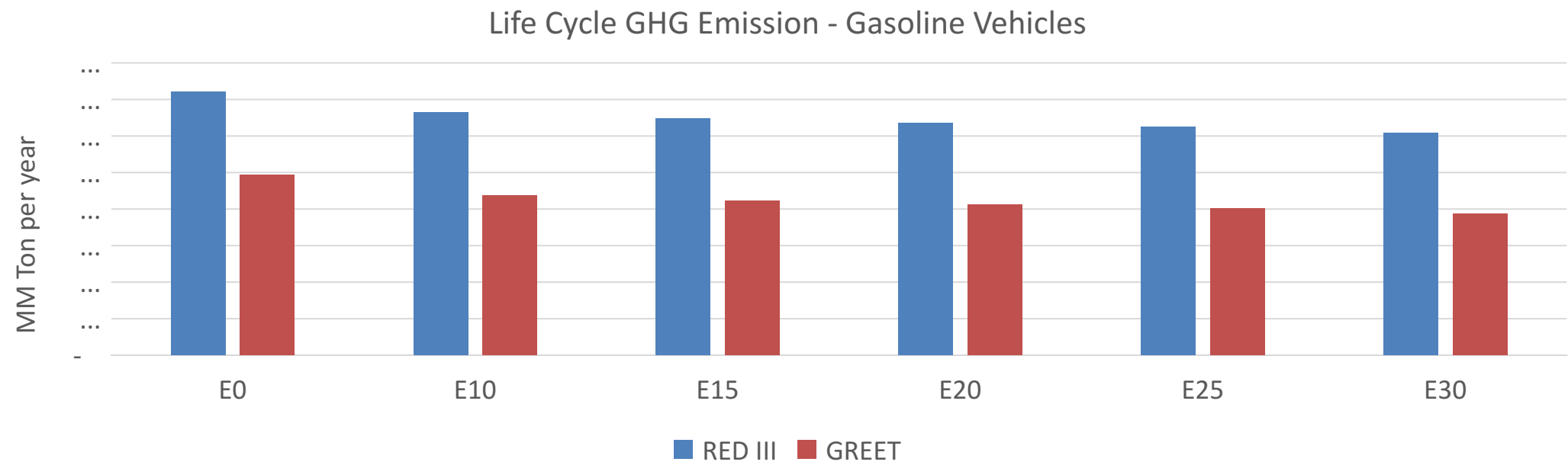
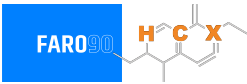
Slovakia – Vehicular Emissions



Slovakia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	99,865	94,329	88,793	84,675	80,722	77,741	74,129	-11%	-19%
CO2	9,214,229	8,977,292	8,753,517	8,577,526	8,490,438	8,440,370	8,284,790	-5%	-8%
VOC	10,576	10,223	9,870	9,671	9,511	9,401	9,210	-7%	-10%
NOx	9,439	7,079	6,608	6,228	5,873	5,482	5,069	-30%	-38%
SOx	104	96	88	81	75	68	61	-15%	-28%
NH3	3,205	3,174	3,142	3,157	3,192	3,217	3,238	-2%	0%
N2O	103	102	102	103	105	106	107	-1%	2%
CH4	2,586	2,586	2,586	2,625	2,681	2,731	2,777	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	109	106	103	102	102	101	100	-5%	-7%
Acetaldehyde	233	267	392	597	813	929	1,098	68%	249%
Formaldehyde	744	724	844	991	1,038	1,148	1,250	13%	39%
Benzene	187	179	170	167	166	159	153	-9%	-11%
PM2.5	590	524	458	398	338	274	208	-22%	-43%
PM10	403	358	313	272	231	187	142	-22%	-43%

Slovakia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	36.8
Target @ 2035 (MMT/year)	20.7
Delta	16.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	11.2%	14.2%	16.5%	18.6%	21.6%
RED II/III	0.0%	8.1%	10.3%	12.0%	13.5%	15.7%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.4%	4.4%	5.0%	5.7%	6.6%
RED II/III	0.0%	3.6%	4.6%	5.4%	6.1%	7.0%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90